Lakshay

+91 9354036350 | lakshayp95@gmail.com | LinkedIn | GitHub

Technical Skills

Languages: C/C++, JavaScript, SQL

Frontend: HTML5, CSS3, Tailwind CSS, React.js

Backend: Node.js, Express.js, REST APIs, LLM API Integration

Authentication & Security: JWT, OAuth 2.0, RBAC

Databases: MongoDB, PostgreSQL, Redis

Tools: Git, GitHub, Postman, Vercel, Render

Projects

CodePilot | AI-Powered Code Analysis Platform
Full Stack, LLM APIs | GitHub

- Built a full-stack AI-powered code assistant that automates debugging, optimization, code review, and code analysis through a unified interface.
- Developed **RESTful API endpoints** to integrate **LLM services** with the frontend, handling code submissions and action-specific analysis requests.
- Implemented **JWT-based authentication** and **protected routes** to secure user-specific resources and prevent unauthorized access
- Integrated **7 AI-assisted workflows** including code explanation, debugging, optimization, review, complexity analysis, quality assessment, and automated test-case generation.

Blink | Scalable URL Shortening Service
React, Node.js, MongoDB, Redis | GitHub

- Designed a URL shortening service that enables users to create, manage, and monitor shortened links through a centralized dashboard.
- Implemented a **Redis cache-aside layer**, reducing average redirect response latency by **49%** while doubling the application throughput.
- Optimized MongoDB lookups using strategic **index-based queries**, improving database throughput by **46%**.
- Integrated **JWT authentication, Google OAuth 2.0, and API rate limiting** to secure user access and protect APIs against abuse and brute-force attempts.

Experience

DRDO – Solid State Physics Laboratory (SSPL) June 2026 – July 2026
Summer Intern | Aerial Object Detection Workflow Analysis

- Analyzed the end-to-end **YOLOv8-based aerial object detection pipeline**, tracing image, video, and webcam inputs through detection, threat classification, and alert generation.
- Evaluated model performance using **precision (88.7%), recall (85.0%), and mAP@50 (90.5%)**, examining class-level results and potential domain gaps.
- Investigated system workflow consistency across **SAFE, MONITOR, and SUSPICIOUS** classifications, CSV logging, snapshot capture, and webhook-based alerts.

Education

Bhagwan Parshuram Institute of Technology August 2023 – July 2027
B.Tech in Electronics and Communication Engineering *CGPA: 8.43*

Convent of Gagan Bhart School, Uttam Nagar, New Delhi April 2022 – March 2023
Class XII, CBSE *81%*

Convent of Gagan Bhart School, Uttam Nagar, New Delhi April 2020 – March 2021
Class X, CBSE *78.8%*